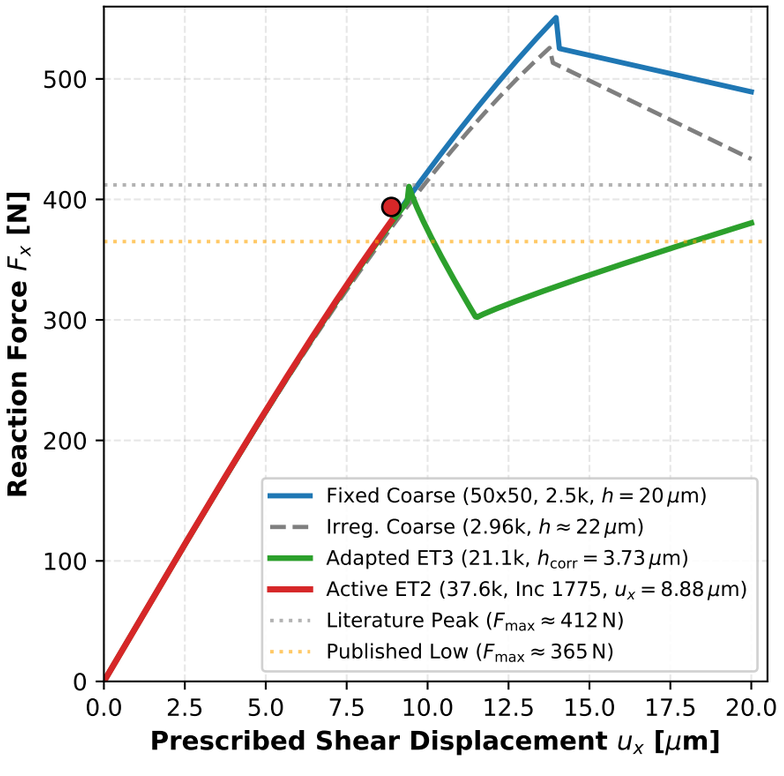
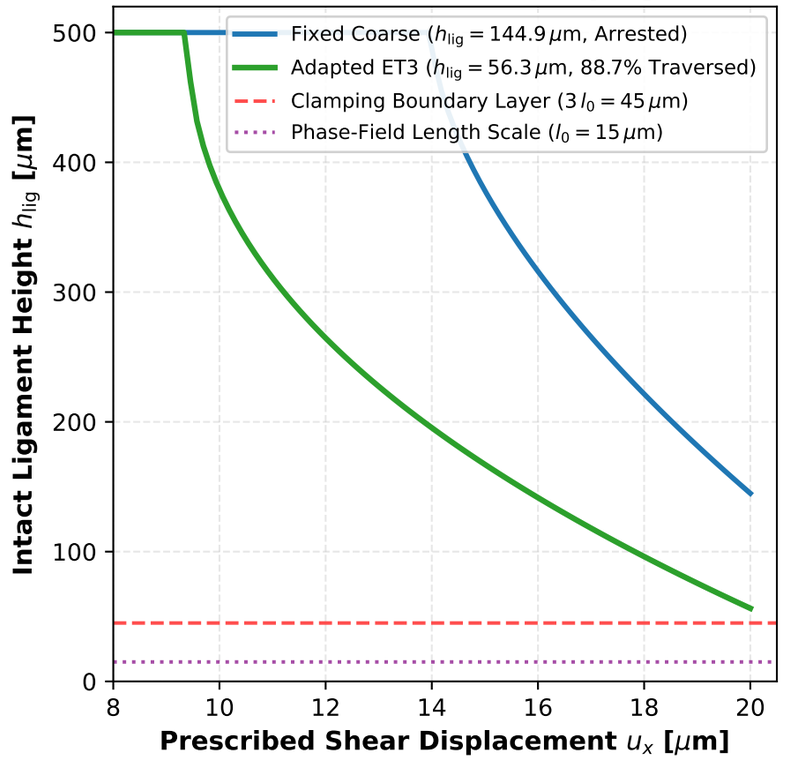
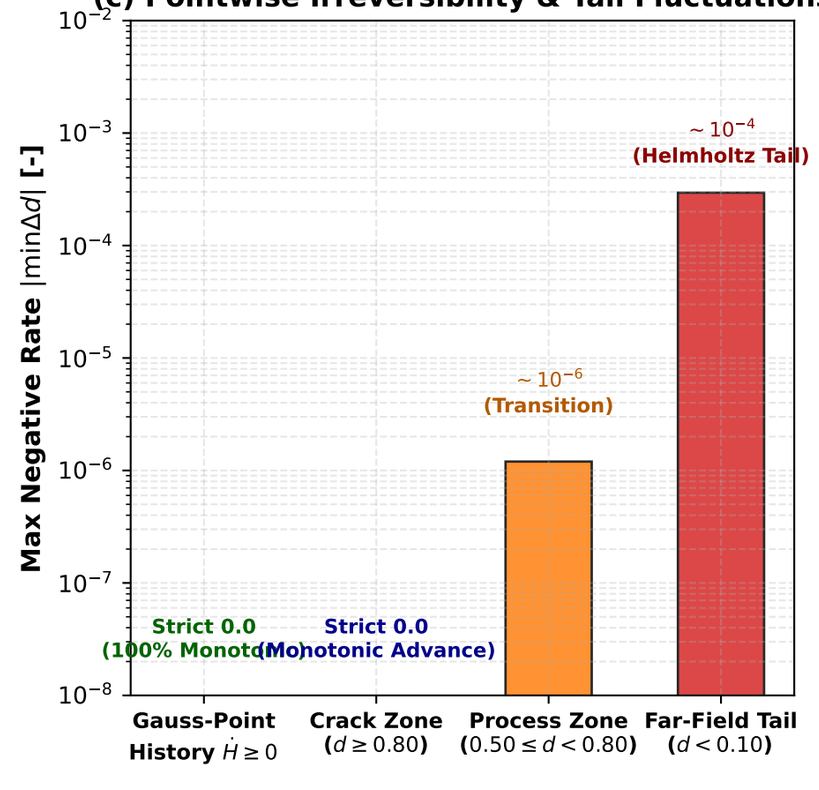
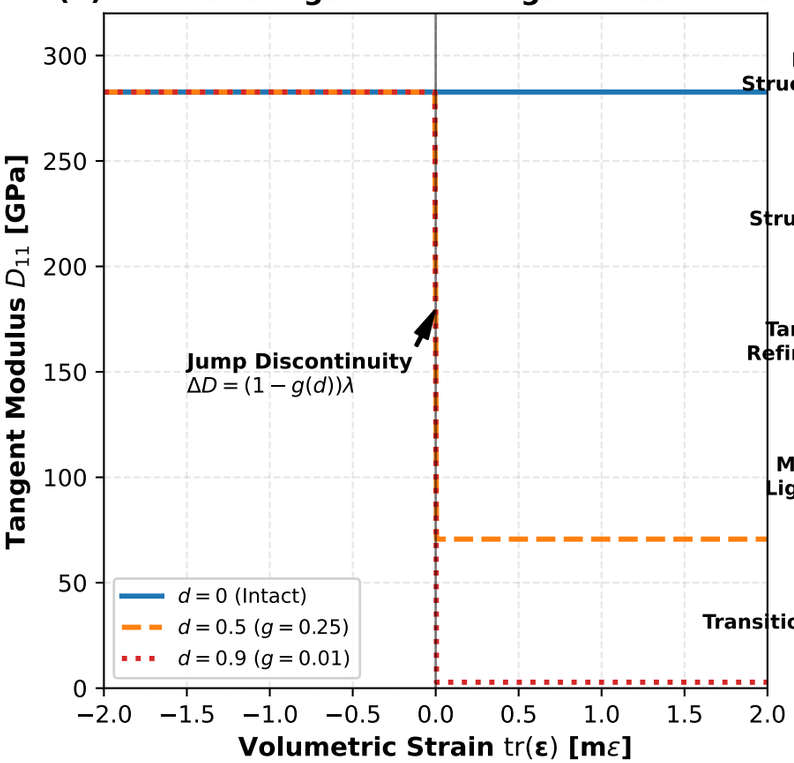
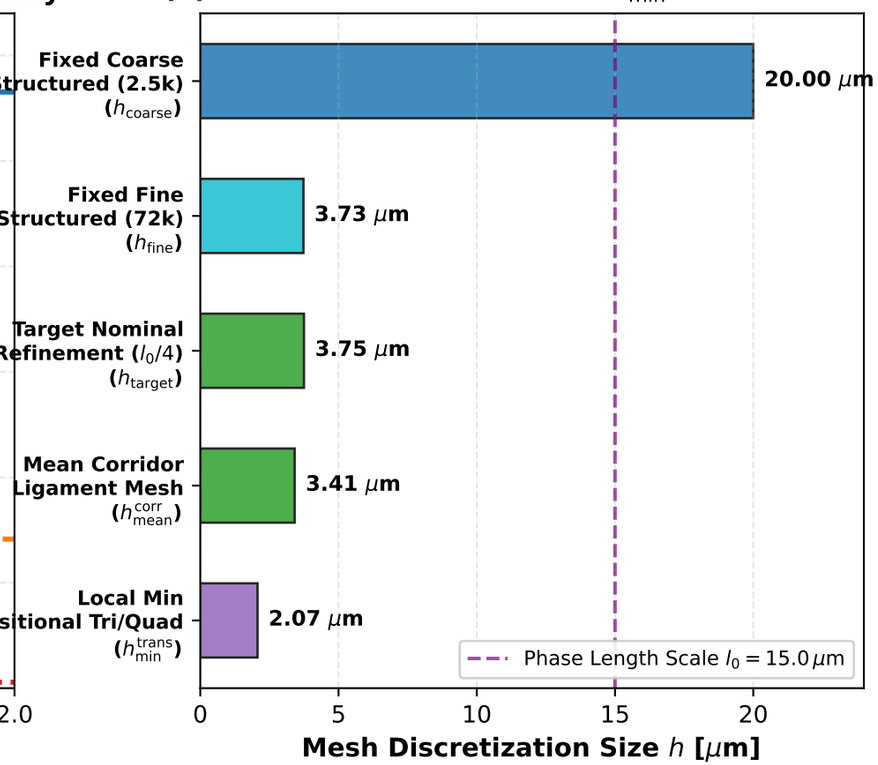
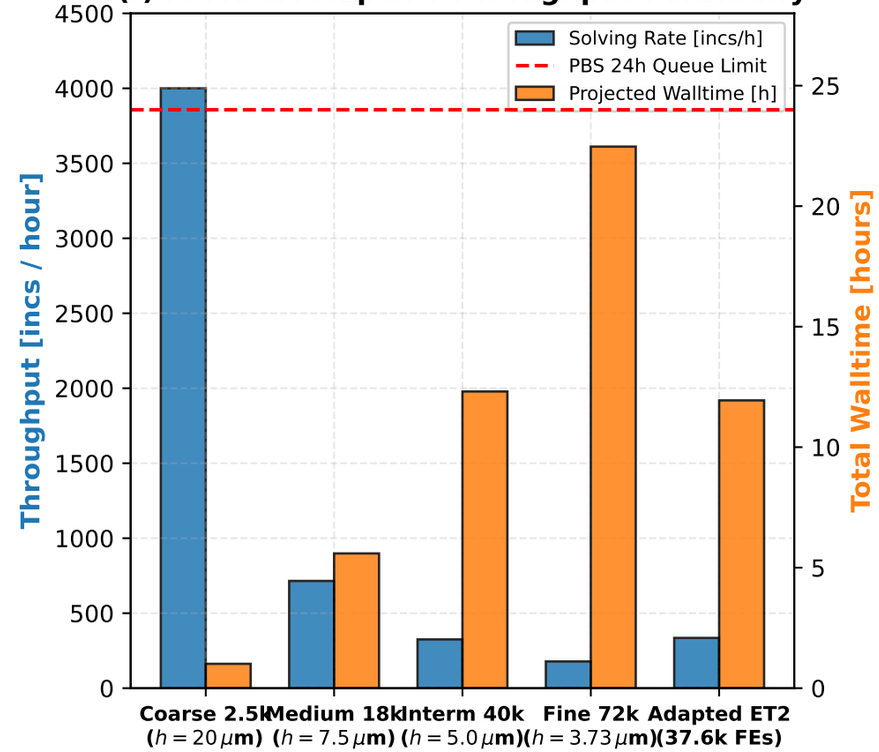

(a) Full Monotonic Mode-II Response ($F-u$)
Reaction Force F_x [N]
Prescribed Shear Displacement u_x [μm]
Fixed Coarse (50x50, 2.5k, $h=20\,\mu$m)
Irreg. Coarse (2.96k, $h\approx22\,\mu$m)
Adapted ET3 (21.1k, $h_{corr}=3.73\,\mu$m)
Active ET2 (37.6k, Inc 1775, $u_x=8.88\,\mu$m)
Literature Peak ($F_{max}\approx412$ N)
Published Low ($F_{max}\approx365$ N)


(b) Crack Penetration & Clamping Arrest
Intact Ligament Height h_{lig} [μm]
Prescribed Shear Displacement u_x [μm]
Fixed Coarse ($h_{lig}=144.9\,\mu$m, Arrested)
Adapted ET3 ($h_{lig}=56.3\,\mu$m, 88.7% Traversed)
Clamping Boundary Layer ($3\,l_0=45\,\mu$m)
Phase-Field Length Scale ($l_0=15\,\mu$m)


(c) Pointwise Irreversibility & Tail Fluctuations
Max Negative Rate |minΔd| [-]
~ 10^{-4}
(Helmholtz Tail)
~ 10^{-6}
(Transition)
Strict 0.0
(100% Monotonic)
Strict 0.0
(Monotonic Advance)
Gauss-Point History $\dot{H}\geq0$
Crack Zone ($d\geq0.80$)
Process Zone ($0.50\leq d<0.80$)
Far-Field Tail ($d<0.10$)


(d) 2D Miehe Algorithmic Tangent Discontinuity
Tangent Modulus D_{11} [GPa]
Volumetric Strain tr($\boldsymbol{\varepsilon}$) [mε]
Jump Discontinuity
$\Delta D=(1-g(d))\lambda$
$d=0$ (Intact)
$d=0.5$ ($g=0.25$)
$d=0.9$ ($g=0.01$)


(e) Discretization Metrics & h_{min} Reconciliation
Fixed Coarse Structured (2.5k) (h_{coarse})
20.00 μm
Fixed Fine Structured (72k) (h_{fine})
3.73 μm
Target Nominal Refinement ($l_0/4$) (h_{target})
3.75 μm
Mean Corridor Ligament Mesh (h^{corr}_{mean})
3.41 μm
Local Min Transitional Tri/Quad (h^{trans}_{min})
2.07 μm
Phase Length Scale $l_0=15.0\,\mu$m
Mesh Discretization Size h [μm]


(f) Fixed & Adaptive Throughput Feasibility
Solving Rate [incs/h]
PBS 24h Queue Limit
Projected Walltime [h]
Throughput [incs / hour]
Total Walltime [hours]
Coarse 2.5k ($h=20\,\mu$m)
Medium 18k ($h=7.5\,\mu$m)
Interm 40k ($h=5.0\,\mu$m)
Fine 72k ($h=3.73\,\mu$m)
Adapted ET2 (37.6k FEs)